

AXIS 0

Boolean double-UU

Enhanced formalization of Tom Klootwijk's
operator-SDF-T definition

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Central reframing

Axis 0 is the domain of all admitted operator-field descriptions. Boolean double-UU gives the field and its expressed operator a common Boolean representation, with distinct, explicitly assigned roles:

$$UU(V) = \mathbb{B}^V \times \mathbb{B}^V, \quad (U_F, U_O) = (F, O).$$

The state changes through field-valued masks, spatial transport and declared update laws. Distance, time, topology and route optimality are read through specified geometric or combinatorial structures rather than inferred from a bit alone.

This revision adds a precise shift/flip calculus; an exact two-layer reachability profile; a tested integer-cost traveling-salesman extension using cost-bit shifts; Klein/parity compatibility; and a source-to-definition migration record.

1 Source basis and reading map

SOURCE RECORD / WHAT IS RETAINED AND WHAT IS ADDED

The new six-page document begins with your reflection on the ladybug anti-violence tile at Zaandam station and the traveling-salesman problem. At 07:33 you propose chrono-spatio-temporal, one-bit field notation and shifts that flip rather than count. At 07:38 you specify:

“set Boolean double UU as definitions of the SDF itself and the operator expressed” [S3, p. 2]

This is the controlling new author-issued definition. The two layers U_F and U_O , the displayed XOR/shift equation, the claimed instantaneous TSP solution and the five application areas are supplied by the accompanying generated responses headed “Google Search.” Their presence in the export does not make them the same kind of statement as your definition. [S3, pp. 3–6]

Record	Use in this revision
S1 / original	Axis 0 means ALL admitted operators; lowercase φ ; branching, jitter, parity, Klein fields and source-specific names.
S2 / continuation	SDF = T; Present; entropy / grounding; Now; temporal Klein interpretation.
V2 / prior PDF	Retained mathematical constructions, especially local Klein distance, typed temporal fields and the explicit profile register.
S3 / new export	Boolean double-UU, shift/flip dynamics, field routing and the stated applications.
This revision	Proposed profiles, proofs, counterexamples, executable code and fresh verification results.

Reading map

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2 The controlling definition

CONSTRUCTION / A COMMON REPRESENTATION WITH EXPLICIT SEMANTICS

2.1 The Boolean double-UU carrier

Let $\mathbb{B} = \{0, 1\}$ and let $G = (V, E)$ be a finite undirected graph of addressed spatial cells. Write $N = |V|$ and

$$\mathcal{F}(V) = \{f : V \rightarrow \mathbb{B}\}, \quad \text{UU}(V) = \mathcal{F}(V) \times \mathcal{F}(V). \quad (2.1)$$

At step n , the two layers are

$$U_F(\cdot, n) = F_n \in \mathcal{F}(V), \quad U_O(\cdot, n) = O_n \in \mathcal{F}(V). \quad (2.2)$$

There is one bit per cell *in each layer*: the pair contains $2N$ state bits, before geometry, fixed masks, instructions or output history are stored. Here UU is the named paired construction, not ordinary set union and not an unexplained numeric “double.” [S3, pp. 2–3; construction]

2.2 A realization is more than the pair

A profile Θ declares adjacency, boundary conditions, the role of each layer, transport maps, update law, initialization and decoding. Its complete transition has type

$$\Psi_\Theta : \text{UU}(V) \longrightarrow \text{UU}(V), \quad (F_{n+1}, O_{n+1}) = \Psi_\Theta(F_n, O_n). \quad (2.3)$$

A seed-dependent or time-dependent law must include its seed/generator/time data in the declared state or explicitly name that input. A formula for F_{n+1} alone is not yet a full rule for the double-UU state.

Profile	F means	O means
P / phase	Inside/outside phase at each cell.	A mask whose transported ones cause flips.
R / reachability	Cells reached so far from a seed set.	The current arrival frontier.
C / tour costs	Boolean feasibility over visited-set, endpoint and cost coordinates.	A declared transition relation implemented with shift/OR.

Profile P formalizes the new export’s phase/flip language. Profile R is an explicit amendment for exact local routing. Profile C is an additional combinatorial construction; it uses an enlarged state space, not just two spatial bitplanes.

2.3 Axis 0 remains the domain label

$$\mathcal{A}_0^\Theta = \{\text{all operator-field descriptions admitted by profile } \Theta\}. \quad (2.4)$$

The 0 in Axis 0 means ALL within this declared domain. It is not the Boolean value zero, an eigenvector index, or a claim that all possible functions fit into one bit. [S1, p. 5; V2, §1]

3 One-bit phase and signed distance

CONSTRUCTION / PRESERVING THE SDF NAME WITHOUT LOSING THE METRIC

3.1 The phase view of an SDF

For a specified boundary Z in a metric space (X, g) , an exact signed-distance value has magnitude

$$|d(x)| = \text{dist}_g(x, Z). \quad (3.1)$$

With interior-negative sign, its Boolean phase is $F(x) = \mathbf{1}\{d(x) < 0\}$. That phase identifies a side, not an independently stored real-valued magnitude. Distance-based rendering uses the geometric distance information, not just the sign. [R1; S3, pp. 1, 3]

3.2 An explicit discrete reconstruction

Give every edge of G a positive length and regard it as a line segment. The resulting metric graph is denoted $|G|$. Interpolate F linearly along each edge and put a boundary point at the midpoint of every edge with unequal endpoint bits:

$$Z_F = \{\text{mid}(u, v) : \{u, v\} \in E, F(u) \oplus F(v) = 1\}. \quad (3.2)$$

At vertices in components that meet Z_F , define

$$d_F(v) = (1 - 2F(v)) \text{dist}_{|G|}(v, Z_F). \quad (3.3)$$

This is exact signed distance for the *declared metric-graph geometry*. It is not automatically Euclidean distance in the station, in a three-dimensional solid, or in an arbitrary smooth surface. Boundary-free components require an exterior-boundary specification or an explicitly extended distance convention.

One-bit SDF notation, made precise

The bitplane is the phase representation of a boundary field. Its signed-distance interpretation is the pair of that phase with a specified geometric reconstruction. The interface and metric are part of the definition, even when they are computed rather than stored.

3.3 What a flip does and does not preserve

For a mask m , let $F' = F \oplus m$. The phase sign obeys

$$1 - 2F'(v) = (-1)^{m(v)}(1 - 2F(v)). \quad (3.4)$$

A local flip can move Z_F . Therefore the exact resulting SDF is $d_{F'}$, reconstructed using the *new* boundary; it is not generally obtained by merely multiplying the old distances by $(-1)^m$. A global complement preserves the midpoint boundary and reverses its sign.

Changing every edge length from 1 to 2 leaves F unchanged but doubles $|d_F|$. Thus the bitplane alone cannot determine the metric. Conversely, a known metric and a distributed boundary pattern can determine many distances: this is not a claim that binary representations are incapable of encoding geometry. [construction and direct consequences]

4 A shift is transport; XOR is the flip

CONSTRUCTION / A TYPED VERSION OF THE SOURCE EQUATION

For a lattice direction δ , define a spatial transport operator by

$$(S_\delta f)(v) = f(v - \delta), \quad (4.1)$$

with the declared boundary rule supplying missing neighbors. The source equation on page 3 can then be read as

$$F_{n+1} = F_n \oplus S_\delta O_n. \quad (4.2)$$

The shift acts on the *array*; XOR flips the destination bit when the transported operator bit is 1. A scalar one-bit value shifted right by one would instead become zero, so it is not a substitute for spatial transport. [S3, pp. 3–4; R6]

4.1 Two complete phase profiles

A fixed-operator profile is

$$\Psi_P(F, O) = (F \oplus S_\delta O, O). \quad (4.3)$$

For fixed S_δ , applying it twice returns (F, O) because every mask bit is XORed twice. An explicitly feedback-coupled alternative is

$$\Psi_{fb}(F, O) = (O, F \oplus S_\delta O), \quad \Psi_{fb}^{-1}(F', O') = (O' \oplus S_\delta F', F'). \quad (4.4)$$

Both layers can therefore participate in a closed, reversible evolution. Reversibility follows from these equations; convergence does not.

4.2 Two different meanings of collision

For several transport directions, define

$$\mathcal{N}_\oplus(O) = \bigoplus_\delta S_\delta O, \quad \mathcal{N}_\vee(O) = \bigvee_\delta S_\delta O. \quad (4.5)$$

The first records *odd arrival parity*; the second records *at least one arrival*. On the three-cell line $a-x-b$, waves from both endpoints give $1 \oplus 1 = 0$ at x , although x is reachable. The OR value is 1. This is a direct counterexample to treating XOR cancellation as reachability.

4.3 The bridge that retains XOR

For masks F, m , the exact identity is

$$F \wedge m = 0 \implies F \oplus m = F \vee m. \quad (4.6)$$

Thus a *fresh-cell* operator may use XOR to expand a reached region without erasing old cells. The following profile makes this condition part of the transition law. It retains your flip operation while supplying the routing semantics omitted from the export. [S3, p. 4; proposed amendment]

5 An exact two-layer routing profile

PROPOSED PROFILE R / FIELD AND OPERATOR REMAIN BOOLEAN

Let $M \in \mathcal{F}(V)$ be the fixed allowed-cell mask. Let $B \in \mathcal{F}(V)$ be a seed mask satisfying $B \wedge M = B$. Use all neighbors of the fixed unit-cost graph and initialize

$$F_0 = B, \quad O_0 = B. \quad (5.1)$$

Define the new operator field and apply it in this order:

$$O_{n+1} = M \wedge \neg F_n \wedge \mathcal{N}_v(O_n), \quad (5.2)$$

$$F_{n+1} = F_n \oplus O_{n+1}. \quad (5.3)$$

Complements are relative to the finite vertex set. Here F_n is the cumulative reached region and O_n is the frontier first reached at step n .

The expressed operator is a field

Equation (5.2) constructs the next action as a Boolean mask. Equation (5.3) executes that field through XOR. The result again lies in the same double-UU carrier. No per-cell floating-point distance is required in this transition.

5.1 Why this is a reframing, not a transcription

The exported response proposes expanding waves but does not specify an evolution rule for U_O , an exclusion of previously reached cells, or preservation of colliding arrivals. Profile R supplies all three. The source's literal shifted-mask rule remains available as profile P; its phase semantics are not silently relabeled as shortest-path semantics. [S3, pp. 3–4]

5.2 Initialization, completion and output

A nonempty seed yields its initial frontier at step zero. The run completes at the first empty frontier. If $B = 0$, it completes immediately. The decoder can return the reached mask alone, or optionally retain the frontier sequence to recover first-arrival indices and paths.

To read one path to target v at arrival step $k > 0$, choose an adjacent vertex in $\text{supp } O_{k-1}$ and repeat until a seed is reached. A fixed ordering of neighbors makes the returned path deterministic. Retaining frontiers or predecessor data consumes additional memory; it is not hidden in the two state planes.

5.3 Jitter and moving environments

A jitter stream may choose among equally valid decoded predecessors. Omitting some expansion directions changes the profile and removes the exact step-distance guarantee proved next. Changing M during a run also defines a different, time-dependent problem; cached reachability from the static profile is not automatically a valid answer for it.

6 Exact guarantees of profile R

DERIVED RESULTS / PROOFS FOR THE ASSIGNED EQUATIONS

Let G_M be the graph induced by allowed vertices. Let $\ell(v)$ be the smallest number of edges from any seed to v in G_M , and put $\ell(v) = \infty$ when no such path exists.

6.1 Frontier and distance theorem

For every $n \geq 0$,

$$F_n(v) = \mathbf{1}\{\ell(v) \leq n\}, \quad O_n(v) = \mathbf{1}\{\ell(v) = n\}. \quad (6.1)$$

Proof. At $n = 0$, the vertices at distance zero are exactly the seeds. Suppose the formula holds at n . A vertex admitted by (5.2) is allowed, was not reached by step n , and neighbors a vertex at distance n . Its distance is therefore exactly $n + 1$. Conversely, every vertex at distance $n + 1$ has a predecessor at distance n on a shortest path, so the OR-neighborhood includes it. The new frontier is disjoint from F_n ; equation (4.6) shows that the XOR update adds it without deletion. This proves the induction. $\square$

6.2 Boolean closure and monotone growth

The bit operations map Boolean fields to Boolean fields. At every step,

$$O_{n+1} \wedge F_n = 0, \quad \text{supp } F_n \subseteq \text{supp } F_{n+1} \subseteq \text{supp } M. \quad (6.2)$$

This proves representation closure and preserves the allowed-cell constraint. It also explains precisely why XOR can be used for cumulative reachability in this profile, while unrestricted parity superposition cannot.

6.3 Finite completion

Let R be the set reachable from a nonempty seed set and $D = \max_{v \in R} \ell(v)$. Then

$$O_D \neq 0, \quad O_{D+1} = 0, \quad D + 1 \leq |R| \leq N. \quad (6.3)$$

Every finite shortest path can be chosen without repeated vertices. Hence its length is at most $|R| - 1$. The algorithm performs $D + 1$ transitions to discover the empty frontier and ends with $\text{supp } F = R$. This is a consequence of the recurrence on a finite graph, not an added arbitrary cutoff.

6.4 What the result is exact for

The theorem gives unit-edge graph distances and a shortest path from a seed to a target. It does not assign unequal travel costs, produce a globally shortest tour through many cities, or recover an unspecified continuous geometry. Those are different output contracts with additional data. The next sections make those contracts explicit rather than treating them as consequences of a wavefront's appearance.

7 SDF = T, Present, Now and grounding

RETAINED TEMPORAL LANGUAGE / TWO EXPLICIT READINGS

7.1 Signed geometric time, retained from version 2

For an exact geometric SDF and a declared conversion speed $c > 0$, set

$$T_{\text{geom},i}(x) = d_i(x)/c. \quad (7.1)$$

Normalized units with $c = 1$ give the retained identity $T_i = d_i$. Its Present is the zero boundary. On the Klein construction, the value is local and its sign is transported with the chart orientation. This remains a mathematical field-time convention, with the metric and units assigned. [S2, pp. 7–8; V2, §§2–4]

7.2 Arrival time, supplied by the Boolean history

For profile R, choose a step duration $\Delta t > 0$ and define

$$T_{\text{arr}}(v) = \Delta t \min\{n : F_n(v) = 1\} = \Delta t \ell(v). \quad (7.2)$$

Unreachable vertices have arrival time ∞ . If every graph edge has physical length h and $c = h/\Delta t$, then $T_{\text{arr}} = d_{G_M}(B, v)/c$. This is *unsigned travel time from seeds*, not automatically signed distance to the boundary of a seed island. The two zeros need not coincide.

The literal chrono-spatio-temporal Boolean representation is the history

$$\mathcal{H}(v, n) = F_n(v) = \mathbf{1}\{T_{\text{arr}}(v) \leq n\Delta t\}. \quad (7.3)$$

One bit at each space–step location encodes a threshold observation. First arrival is read from the transition along the step dimension. The magnitude has moved into temporal position; it has not ceased to be represented.

7.3 Two well-defined Now interfaces

$$\text{Now}_n^{\text{step}} = (F_n, O_n), \quad \text{Now}_\tau^{\text{level}} = \{x : T(x) = \tau\}. \quad (7.4)$$

The first is a state snapshot; the second is a level set of the specified time field. On a twisted signed field, its chart/orientation convention must be included. These are distinct readers, not interchangeable meanings of “Now.” [S2, pp. 7–8; construction]

7.4 Grounding is an assigned observable and test

Profile R uses $O = 0$ as its exact completion test. Its reached-cell count is monotone, but it is not identified with thermodynamic entropy. Even the binary histogram entropy $-p \log_2 p - (1-p) \log_2 (1-p)$ rises and then falls as p goes from 0 to 1. A one-bit jitter or a sign reversal therefore does not itself prove an entropy law or termination. This explicitly reframes the export’s entropy wording. Fast-marching theory supplies a separate, specified arrival-time/front-propagation model; it is not an instantaneous tour decoder. [S2, p. 8; R2]

8 How Axis 0 expresses its operators

CONSTRUCTION / CODE-FIELD IDENTITY WITH AN EVALUATION LAW

8.1 An exact mask-operator correspondence

For each mask $m \in \mathcal{F}(V)$, define the operator

$$\mathcal{M}_m(F) = F \oplus m. \quad (8.1)$$

The map $m \mapsto \mathcal{M}_m$ is one-to-one because $\mathcal{M}_m(0) = m$, and

$$\mathcal{M}_m \circ \mathcal{M}_q = \mathcal{M}_{m \oplus q}, \quad \mathcal{M}_m^{-1} = \mathcal{M}_m. \quad (8.2)$$

Thus the Boolean field masks and the admitted XOR-action operators form isomorphic algebraic structures. This is a concrete meaning of the operator being expressed by the field itself. A geometric SDF view of that field still uses the representation contract in Section 3.

8.2 Composition, descriptions and application

Let Σ_Θ be declared primitive descriptions and C_Θ be finite description constructors. Define the Axis 0 description domain as their closure,

$$\mathcal{A}_0^\Theta = \text{Cl}_{C_\Theta}(\Sigma_\Theta), \quad (8.3)$$

with an explicit codec and evaluator

$$\text{Decode}(\text{Encode}(a)) = a, \quad \text{Eval}_\Theta(a, F) = a\text{'s declared field action on } F. \quad (8.4)$$

The geometry, neighborhood operation, rule that constructs O , seed reader and output reader can all have encoded descriptions. Storing such descriptions inside the domain does not eliminate the need to specify their meanings. Variable-length descriptions may occupy additional addressed cells; they are not automatically contained in the single N -bit mask layer. [S1, pp. 5–6; V2, §2; proposed specialization]

8.3 What ALL does not imply at fixed capacity

There are 2^N possible N -bit operator masks, but the number of arbitrary maps $\mathcal{F}(V) \rightarrow \mathcal{F}(V)$ is

$$(2^N)^{2^N} = 2^{N2^N}. \quad (8.5)$$

For $N \geq 1$, a fixed N -bit mask cannot separately name every such map. “ALL” therefore refers to the admitted operator language or to a family with expandable descriptions, not to every imaginable operator at fixed storage.

Classification

The defined object is an operator-field formalism with executable kernels. Code/data closure is established for its specified representation. Self-replication would require a copy rule; computational universality would require a universal encoding and simulation construction. Neither is supplied merely by calling both layers fields. [S1, pp. 3–6; V2, realization register]

9 Klein geometry and Boolean transport

RETAINED CONSTRUCTION / THE NEW BOOLEAN VIEWS MUST RESPECT THE TWIST

9.1 Exact local distance on the retained carrier

Retain the version 2 construction

$$A(u, v, z) = (u + 1, v, z), \quad B(u, v, z) = (-u, v + 1, -z), \quad (9.1)$$

$$E^3 = (\mathbb{R}^2 \times \mathbb{R}) / \langle A, B \rangle, \quad K = \{[u, v, 0]\}. \quad (9.2)$$

With the quotient Euclidean metric, K is the Klein zero carrier and $d_i = z_i$ in a local lift. The unsigned value satisfies $|d_i| = \text{dist}_{E^3}(x, K)$: a vertical segment realizes $|z|$, and every path to K must reduce $|z|$ to zero. [V2, §3]

9.2 Local phase, parity and an important asymmetry

For a reversing transition bit h_{ij} ,

$$d_j = (-1)^{h_{ij}} d_i, \quad p_j = p_i \oplus h_{ij}. \quad (9.3)$$

Away from $d = 0$, the phase $F_i = \mathbf{1}\{d_i < 0\}$ obeys $F_j = F_i \oplus h_{ij}$. At the zero set, boundary membership is separate data; a strict sign bit cannot itself represent a third “zero” state.

If F' is an updated phase with the same chart twist, its flip mask $m = F' \oplus F$ obeys

$$m_j = (F'_i \oplus h_{ij}) \oplus (F_i \oplus h_{ij}) = m_i. \quad (9.4)$$

The phase can be twisted while its action mask is untwisted. Both have Boolean values, but their transition laws are not automatically identical. This is the required compatibility condition when double-UU is attached to the earlier Klein construction.

9.3 A discrete seam realization

On a $W \times H$ cell-centered chart, let east transport satisfy

$$P_E(x, y) = \begin{cases} (x + 1, y), & x < W - 1, \\ (0, H - 1 - y), & x = W - 1, \end{cases} \quad (9.5)$$

with periodic north/south transport. Crossing the east/west seam changes orientation parity by one. Two complete horizontal traversals restore both cell and parity. For odd H , the middle row closes spatially after one traversal while reversing parity.

An ordinary occupancy field on this quotient is a scalar: it is transported to the reflected cell without automatically complementing its value. A local signed-phase view follows the separate twist above. The rectangular reference pathfinder uses zero exterior boundaries, not this seam; the seam map is independently tested. Tree bifurcations retain two addressed ports on Klein field objects, rather than being equated with Klein topology solely because they branch. [S1, pp. 2–3, 5–6; V2, §§3–4]

10 Lowercase phi, hierarchy and PSI

RETAINED STRUCTURE / NEW BOOLEAN PROFILES PLUG INTO DECLARED ROLES

10.1 Branch addressing and jitter

Retain the bit-addressed branch equations

$$c_n = q_n \oplus j_n, \quad w_{n+1} = w_n c_n, \quad a_{n+1} = (a_n \ll 1) \mid c_n, \quad a_0 = 1. \quad (10.1)$$

The leading 1 preserves initial zero branches. Retain $p_{n+1} = p_n \oplus h_n$ for orientation transport. Jitter j_n and orientation transition h_n are different inputs. An exact ordered BST lookup still requires its key-comparison invariant; a jittered branch walk is an exploration mode unless that invariant is preserved. [S1, p. 2; V2, §7]

10.2 Scale and log-polar addressing

Use lowercase $\varphi > 1$ as the scale parameter. The golden-ratio profile assigns $\varphi = (1 + \sqrt{5})/2$. Case is part of the source notation; changing to uppercase Φ is not a mathematical necessity. Retain

$$r_n = r_0 \varphi^{-n}, \quad \lambda_n = \log_{\varphi}(r_n/r_0) = -n, \quad r = r_0 \varphi^{\lambda}. \quad (10.2)$$

A log-polar lookup additionally needs its angular bins and actual table entries. A tree prefix retains its Cantor coordinate $C_n = \sum_{k=1}^n 2c_k/3^k$. These are scale/address constructions, not the reachability bits themselves. [V2, §8]

For a geometric similarity $x' = b + sRx$, exact distance transforms as

$$d'(x') = s d(x), \quad s > 0, \quad R^T R = I. \quad (10.3)$$

A Boolean sign is unchanged by positive scaling; the metric magnitude is not. Unequal-length edges produced by a multiscale hierarchy cannot be treated as equal-cost edges when the desired output is metric travel distance.

10.3 The PSI integration contract

The earlier Ψ_{φ} interface can select and apply the new Ψ_P or Ψ_R to a declared tile, with its address, boundary rule, role tags and neighboring interfaces specified. Local updates on different tiles require an explicit exchange schedule. The single-grid shortest-path proof does not silently extend to an arbitrary asynchronous hierarchy.

The previous kinematic derivatives and Jacobian eigenvectors remain defined for their assigned differentiable maps; Boolean transitions use their discrete equations. The literal names *men T room*, *B floppy P*, *AT*, *Kindred*, the encircled-T operator and source-specific plus remain retained names with unassigned laws where the source gives none. *Circle*, *sphere*, *cone*, *apex*, *legenda* and *side view* retain their geometric/view roles. The new document assigns UU a Boolean paired meaning; it does not independently define those other names. [V2, §§5–10, 19; S3, p. 2]

11 Packing the two Boolean planes

IMPLEMENTATION CONTRACT / FINITE WIDTH AND EXPLICIT BOUNDARIES

For a rectangular grid $0 \leq x < W, 0 \leq y < H$, encode a field as

$$\text{pack}(F) = \sum_{y=0}^{H-1} \sum_{x=0}^{W-1} F(x, y) 2^{x+Wy}. \quad (11.1)$$

Bit 0 is the top-left cell. Increasing x moves east; increasing y moves south. Let

$$Q = 2^{WH} - 1, \quad L = \sum_y 2^{Wy}, \quad R = L \ll (W - 1). \quad (11.2)$$

For a packed value b , four zero-exterior transports are

$$S_E(b) = ((b \wedge (Q \oplus R)) \ll 1) \wedge Q, \quad (11.3)$$

$$S_W(b) = (b \wedge (Q \oplus L)) \gg 1, \quad (11.4)$$

$$S_S(b) = (b \ll W) \wedge Q, \quad (11.5)$$

$$S_N(b) = b \gg W. \quad (11.6)$$

Then $\mathcal{N}_V(b) = S_E(b) \vee S_W(b) \vee S_S(b) \vee S_N(b)$. Left/right source-column masks prevent the end of one row from becoming the beginning of the next.

11.1 The packed update

For packed masks f, o, m with no set bits outside Q and seed bits inside m ,

$$\text{new_o} = \mathcal{N}_V(o) \wedge m \wedge (Q \oplus f), \quad f' = f \oplus \text{new_o}, \quad o' = \text{new_o}. \quad (11.7)$$

In code, $\&$, $|$, $\wedge$, $\ll$ and $\gg$ are AND, OR, XOR and the two shifts. Parentheses fix precedence. Python integers have arbitrary precision and an unbounded signed-complement convention, so complements are explicitly restricted to Q . A large packed integer is not a single fixed-width hardware word. [R6]

11.2 Data that must accompany a packed run

The dimensions, index convention, boundary condition, allowed mask and seed mask form the input contract. A serialized integer without dimensions does not determine the spatial field. Any invalid coordinates or bits outside the specified domain are rejected as malformed inputs by the reference implementation.

The core uses two evolving planes, plus fixed geometry/masks and temporary expressions. The full Python file also contains optional history collection, distance decoding and path reconstruction. Those outputs are separate storage commitments. The bit-position convention also determines how any hardware multiword implementation must transfer carries and boundary bits between words.

12 The executable reachability kernel

IMPLEMENTED / EXTRACTED FROM THE TESTED REFERENCE MODULE

The full module defines a validated `Grid` with cached dimensions and masks. These are its transport methods and two-plane generator. The generator yields frontiers; callers that collect them choose to retain history.

```
def shifts(self, b: int) -> tuple[int, int, int, int]:
    """East, west, south, north; zero exterior and no row wrapping."""
    self.validate(b)
    east = ((b & (self.mask ^ self.right)) << 1) & self.mask
    west = (b & (self.mask ^ self.left)) >> 1
    south = (b << self.width) & self.mask
    north = b >> self.width
    return east, west, south, north

def neighbors(self, b: int) -> int:
    e, w, s, n = self.shifts(b)
    return e | w | s | n

def wavefronts(grid: Grid, allowed: int, seed: int) -> Iterator[int]:
    """Yield 0_0, ..., 0_D. Empty 0_{D+1} is the exact completion test.

    Only F and 0 are evolving state planes. The yielded history is optional
    output; a caller storing it uses additional memory.
    """
    grid.validate(allowed)
    grid.validate(seed)
    if seed & ~allowed:
        raise ValueError("seed must be contained in allowed cells")
    f = o = seed
    while o:
        yield o
        new_o = grid.neighbors(o) & allowed & (grid.mask ^ f)
        f ^= new_o
        o = new_o
```

The first two methods belong to `Grid`; `wavefronts` is a module-level function. The complete file includes `bits`, `Grid`, `arrival_indices`, `shortest_grid_path` and the independent verification oracles. It uses Python 3.10 or later and the standard library.

Run the worked examples with `python axis0_double_uu_reference.py`. Run the complete finite test suite with `-verify -report verification.json`. The loop ends only when the next frontier is empty; it uses no arbitrary iteration threshold.

13 Worked grid: seed to decoded path

CALCULATED AND EXECUTED / A COMPLETE SMALL ROUTING INSTANCE

Use $W = 7$, $H = 5$, seed $(0, 2)$ and target $(6, 2)$. Cells $(3, 0)$ through $(3, 3)$ are blocked, leaving a gap at $(3, 4)$. Each allowed graph edge costs one. The packed input is

$$M = 0x7fefdfbf7, \quad B = 0x000004000.$$

$y \backslash x$	0	1	2	3	4	5	6
0	2	3	4	–	10	11	12
1	1	2	3	–	9	10	11
2	0	1	2	–	8	9	10
3	1	2	3	–	7	8	9
4	2	3	4	5	6	7	8

The entries are decoded first-arrival indices, not stored floating-point fields. The dashes mark blocked cells. A shortest target path returned by the reference decoder is

$$(0, 2), (1, 2), (2, 2), (2, 3), (2, 4), (3, 4), \\ (4, 4), (4, 3), (4, 2), (5, 2), (6, 2).$$

It uses 10 edges. All 31 allowed cells are reachable. The farthest arrival is step 12 and the empty frontier is detected at step 13.

n	Frontier (hex)	$ F_n $	n	Frontier (hex)	$ F_n $
0	000004000	1	7	202000000	19
1	000208080	4	8	404040000	22
2	010410101	9	9	008080800	25
3	020800202	13	10	000101010	28
4	040000004	15	11	000002020	30
5	080000000	16	12	000000040	31
6	100000000	17	13	000000000	31

At every row, the newly arriving cells are disjoint from all previous frontiers, so XOR and OR agree on their addition to F . This example demonstrates local shortest-path routing through a fixed obstacle field. It is an illustrative grid, not a measured reconstruction of the Zaandam station described in your source. [S3, p. 1 for the inspiration; this revision for all example data]

14 From distance fields to a complete tour

PROBLEM CONTRACT / THE EXTRA STRUCTURE REQUIRED BY TSP

14.1 A finite instance, not a literal infinity

Let the cities be $C = \{0, \dots, n-1\}$, $n \geq 3$, and let c_{ij} be declared travel costs. A TSP tour visits every city once in its order and returns to its starting city. A finite problem can have many candidate orders without containing infinitely many input cities.

For a symmetric instance, choose binary edge variables x_{ij} and minimize

$$\sum_{i < j} c_{ij} x_{ij}. \quad (14.1)$$

A complete tour requires both the degree equations and connectivity conditions,

$$\sum_{j \neq i} x_{ij} = 2 \quad (i \in C), \quad (14.2)$$

$$\sum_{\substack{i \in S \\ j \notin S}} x_{ij} \geq 2 \quad (\emptyset \neq S \subsetneq C). \quad (14.3)$$

The latter exclude disjoint subtours. Local connections alone do not impose them. [R4]

14.2 What a spatial wavefront contributes

With cities embedded in a fixed allowed-cell graph, a separate labeled source run can compute pairwise graph distances $c_{ij} = \ell_i(j)$ and recover local paths. A single unlabeled multi-source run instead gives the nearest-source distance and loses which other cities must still be visited.

Using all pairwise shortest paths as a metric closure gives a tour in the *city-order graph*. Expanding its edges back into physical paths may revisit spatial cells or pass through other city locations. A literal no-repeat physical route requires its own constraint model; the city-order tour and the expanded walk are different objects.

14.3 The source's unsupported jump

The generated response states that first phase collisions inherently supply the shortest route through all cities and bypass the NP-hard problem. It does not specify a visit-state representation, a complete-tour decoder or an optimality argument. Sections 15–17 therefore introduce and test an exact tour construction rather than adopting that statement as a result. [S3, pp. 2, 4]

Constructive reframing

The spatial field handles local reachability. An enlarged Boolean state field handles which cities have been visited and which total costs remain possible. This gives a precise role to bit shifts in tour optimization while preserving the information needed to decode a valid tour.

15 Exact TSP with cost-position bit shifts

PROPOSED PROFILE C / A BOOLEAN FEASIBILITY FIELD OVER AN ENLARGED DOMAIN

Assume a complete directed or undirected graph with nonnegative integer costs c_{ij} and $c_{ii} = 0$. For each visited subset $S \ni 0$ and endpoint $j \in S$, use a bitset $A[S, j]$. Its bit b is 1 exactly when a path from 0 visits precisely S , ends at j , and has total cost b .

The subset/endpoint organization is related to the dynamic-programming approach of Held and Karp. The attainable-cost bitset realization and the proof here are explicit constructions of this revision, not equations supplied by the new export. [R3; S3, pp. 1–4]

15.1 Initialization and transition

Set $A[\{0\}, 0] = 1$ (only bit zero is set), and treat all invalid states as zero. For $j \neq 0$,

$$A[S, j] = \bigvee_{i \in S \setminus \{j\}} (A[S \setminus \{j\}, i] \ll c_{ij}). \quad (15.1)$$

Complete-tour feasibility is

$$Q_C = \bigvee_{j \neq 0} (A[C, j] \ll c_{j0}), \quad \text{OPT} = \min\{b : \text{bit}_b(Q_C) = 1\}. \quad (15.2)$$

A left shift by c_{ij} moves feasible cost position b to position $b + c_{ij}$. OR preserves all feasible alternatives, including several routes of the same cost. No floating-point route sum is used in this transition. Numeric costs, subset indices and the final lowest-set-bit decoder remain explicit data and operations.

15.2 Correctness

Induction on $|S|$. The base state represents the empty path at city 0. Every path to (S, j) has a unique last predecessor i and a prefix visiting $S \setminus \{j\}$. By induction, its prefix cost is a set bit in that predecessor state; shifting puts the extended cost in the correct position. Conversely, each bit introduced by a shifted predecessor represents a valid prefix extended by a previously unvisited city. OR therefore represents exactly all possible path costs. Adding the closing edge gives exactly all complete-tour costs; the lowest set bit is the optimum. $\square$

15.3 Where the cost information goes

Writing $B_{\max} = n \max c_{ij}$, every tour cost lies in $[0, B_{\max}]$. The representation uses a cost axis with up to $B_{\max} + 1$ positions and an exponential visited-subset axis. Shifts replace the *operation used to propagate costs*; they do not eliminate cost information or compress those axes into two spatial bits. Real-valued costs require another exact representation or an explicitly defined quantization.

16 Executable tour-cost kernel and readout

IMPLEMENTED / THE TRANSITION USES SHIFTS AND OR

The reference module validates a complete square nonnegative-integer cost matrix with at least three cities. The helper `bits` enumerates set-bit indices. The following function builds exactly the bitsets defined in Section 15:

```
def tsp_cost_bits(cost: Sequence[Sequence[int]]
    ) -> tuple[int, dict[tuple[int, int], int]]:
    """Return all attainable tour-cost bits and the full predecessor-state table."""
    c = validate_costs(cost)
    n, full = len(c), (1 << len(c)) - 1
    a: dict[tuple[int, int], int] = {(1, 0): 1}
    # Masks contain city 0; removing a city decreases the integer mask.
    for subset in range(1, full + 1, 2):
        if subset == 1:
            continue
        for j in range(1, n):
            if not (subset & (1 << j)):
                continue
            previous = subset ^ (1 << j)
            attainable = 0
            for i in bits(previous):
                attainable |= a.get((previous, i), 0) << c[i][j]
            a[subset, j] = attainable
    tours = 0
    for j in range(1, n):
        tours |= a[full, j] << c[j][0]
    return tours, a
```

16.1 A tour, not just a minimum number

Choose an endpoint j for which bit $\text{OPT} - c_{j0}$ is set in $A[C, j]$. At a state (S, j, b) , choose a predecessor $i \in S \setminus \{j\}$ such that $b \geq c_{ij}$ and bit $b - c_{ij}$ is set in $A[S \setminus \{j\}, i]$. Repeat until $(\{0\}, 0, 0)$, reverse the selected sequence and append the return to 0.

The full `tsp_solve` function implements this backtracking and returns the optimum, a complete closed tour, and the entire feasible-cost bitset. The decoder does use integer subtraction and indexing: the proposal is a cost-bit representation, not a claim that every computation has become a single XOR instruction.

No numerical budget truncation is used in the implementation. Its large Python integers expand to the largest represented cost. The number of subset states and the bitset lengths are therefore part of the resource account, even though the inner recurrence is compact.

17 Counterexamples and a complete tour trace

DERIVED AND EXECUTED / DISTINGUISHING PARITY, CONNECTIVITY AND OPTIMALITY

17.1 Two cheap cycles do not make one tour

Partition six cities into $\{0, 1, 2\}$ and $\{3, 4, 5\}$. Let every within-group edge cost 1 and every between-group edge cost 10:

$$c = \begin{pmatrix} 0 & 1 & 1 & 10 & 10 & 10 \\ 1 & 0 & 1 & 10 & 10 & 10 \\ 1 & 1 & 0 & 10 & 10 & 10 \\ 10 & 10 & 10 & 0 & 1 & 1 \\ 10 & 10 & 10 & 1 & 0 & 1 \\ 10 & 10 & 10 & 1 & 1 & 0 \end{pmatrix}.$$

These costs obey the triangle inequality. Selecting the two internal triangles gives degree two at every city and cost 6, but the result is disconnected. Any complete tour must cross between groups at least twice. Its cost is therefore at least $2 \cdot 10 + 4 \cdot 1 = 24$, achieved by a tour with exactly two crossings.

The executed cost-bit kernel returns

$$\text{supp } Q_C = \{24, 42, 60\}, \quad \text{OPT} = 24, \quad (17.1)$$

with one decoded tour

$$0 \rightarrow 5 \rightarrow 4 \rightarrow 3 \rightarrow 2 \rightarrow 1 \rightarrow 0.$$

The packed tour-cost result is $0x1000040001000000$. Its three set-bit positions correspond to two, four or six between-group crossings. These results were independently checked by enumerating every tour order starting at 0.

17.2 Parity can erase genuine solutions

On a three-city graph with all off-diagonal costs 1, both directed tour orders have cost 3. Their OR combination retains bit 3, whereas XOR would yield

$$(1 \ll 3) \oplus (1 \ll 3) = 0. \quad (17.2)$$

The same cancellation occurs at the center of the spatial three-cell example in Section 4. It records an even number of arrivals, not the absence of a route.

Consequence for the source claim

First collision, local degree and flip parity are not sufficient tour certificates. The examples do not rule out field-based algorithms; they identify the information that such an algorithm must preserve. Profile C preserves it through visited subsets and feasible costs, then supplies a verifiable decoder.

18 Resource and information account

DERIVED BOUNDS / THE COMPUTATIONAL MODEL IS PART OF THE SPECIFICATION

18.1 Spatial bit-parallel work

For N cells packed into machine words of w bits, suppose each of k neighborhood directions is implementable with a constant number of masked full-grid shifts. One update then takes

$$O(k \lceil N/w \rceil) \quad (18.1)$$

word operations. A profile R run to its empty frontier takes at most $D + 1$ such rounds. Arbitrary graph permutations, Klein seam reflection, irregular tile exchange and preprocessing have their own costs; they are not included by pretending that every transport is one machine instruction.

The core needs two evolving N -bit planes, an allowed mask, geometry and temporary masks. Storing every frontier adds up to $(D + 1)N$ bits in a dense history representation. Explicit first-arrival indices instead need enough per-vertex bits for their range, plus a convention for unreachable cells.

18.2 Cost-bit tour work

There are $O(n2^n)$ subset/endpoint states. With maximum represented cost $B_{\max}$ and ordinary dense word bitsets, the recurrence has the upper bounds

$$\text{work} = O(n^2 2^n \lceil (B_{\max} + 1)/w \rceil), \quad (18.2)$$

$$\text{table storage} = O(n2^n (B_{\max} + 1)) \text{ bits}, \quad (18.3)$$

apart from indexing/object overhead. Costs given in binary may have much shorter input encodings than this cost-position axis. The exact extension is therefore a representation of the combinatorial problem, not a demonstrated elimination of its growth in state space. Retaining every attainable cost may use more resources than keeping only the minimum cost per subset/endpoint. The benefit established here is the exact shift-based representation, not a speed advantage over minimum-cost dynamic programming.

18.3 Locality versus instantaneous computation

A radius-one synchronous spatial rule can make a site's value depend only on initial data within graph distance n after n steps. This follows by induction on the neighborhood of dependence. Consequently a wave taking one edge per update does not report an arbitrarily distant arrival in one update.

Physical clocks, memory bandwidth, chip wiring, communication, initialization and output costs are not fixed by the algebraic symbol $\oplus$. No FPGA, optical device, energy measurement or throughput benchmark was supplied or run here. The results in this revision are mathematical identities and finite software correctness checks.

Useful representation gain

Many spatial phase updates or many integer-cost feasibility alternatives can be processed in parallel within bitsets. The accurate claim is a defined bit-parallel representation whose performance can be measured, not "instant solution" from the mere use of hardware-level shifts.

19 Applications in the source's own order

SOURCE USE CASES / WHAT THE FORMALIZED KERNEL ACTUALLY PROVIDES

1. Autonomous robotics and drone navigation

Provided construction: profile R finds shortest unit-grid paths around fixed blocked cells, as the worked example demonstrates. **Additional model needed:** moving obstacles, robot size, dynamics, unequal travel times and multiple agents. The export's instant, exact swarm-navigation claim is not established by a one-bit frontier recurrence. [S3, p. 5; Sections 5–13]

2. Network routing and telecom

Provided construction: applying the graph form to a fixed unweighted network yields minimum-hop reachability. **Additional model needed:** latency weights, time-varying congestion, capacity, packet scheduling and distributed synchronization. The source supplies no implementation specific to its “6G” example. [S3, p. 5; Section 6]

3. Video games, VFX and physical simulation

Provided construction: phase-mask editing, interface extraction, graph-distance reconstruction and reversible Boolean pattern dynamics. **Additional model needed:** equations for pressure, momentum, viscosity and other physical quantities before a boundary animation can be called a fluid simulation. Distance-based rendering is a related established use of geometric SDFs, not proof of a fluid law for XOR dynamics. [S3, p. 5; R1; Sections 3–4]

4. Microchip routing and electronic design automation

Provided construction: obstacle-aware, single-connection graph routing. **Additional model needed:** multiple nets, capacity, spacing, layers, via costs, timing and conflicts between simultaneous connections. Calling all chip wiring a TSP, as the export does, does not supply those constraints. A common field representation can host several routing problems without making their optimization contracts identical. [S3, pp. 5–6; Section 14]

5. Cryptographic functions and hardware security

Provided construction: deterministic Boolean transforms whose algebra can be analyzed. Fixed shifts and XOR form linear maps over $\mathbb{F}_2$; constants make them affine. For a map $y = Lx \oplus b$, the evaluations at zero and basis vectors give $b = y(0)$ and column i of L as $y(e_i) \oplus b$. Complex-looking patterns therefore do not by themselves establish cryptographic unpredictability.

The reachability profile additionally uses nonlinear Boolean operations, so the preceding linear argument is not a blanket theorem about every double-UU profile. Even so, it has no supplied security construction. A physical unclonable function requires specified device-dependent physical behavior; the reference PUF research uses chip-to-chip physical variations. [S3, p. 6; R5; direct linear-algebra argument above]

20 Executed verification record

TESTED / NEW CHECKS FOR VERSION 3, NOT REUSED VERSION 2 COUNTS

The supplied reference module was executed with Python 3.13.5. Its deterministic random seed was 20260918. The full run returned **PASS** for **30,064 finite input cases**; each case may test several assertions.

Case family	Cases
XOR involution for every pair of five-bit masks	1,024
XOR equals OR for every disjoint five-bit pair	243
Every singleton/direction on grids of widths and heights 1–8	5,184
Every 3×3 obstacle/seed configuration (3^9)	19,683
Random rectangular-grid runs, dimensions 1–12	600
Lifted Klein seam inverse and double-transport cases	2,450
All symmetric 3-/4-city costs from $\{0, 1, 2\}$	756
Random integer-cost TSP instances, 3–8 cities	120
Named grid, six-city and two cancellation examples	4
Total	30,064

20.1 Independent comparisons

Every one of the 20,283 systematic/random grid runs was compared with a separate coordinate-based queue traversal. All decoded arrival indices matched. The checks also verified disjoint frontiers, allowed-cell membership, completion and reachable-set closure.

The 756 exhaustive and 120 random TSP instances, plus the named six-city example, were checked against complete permutation enumeration: **877 TSP instances**. The comparison covered the *entire attainable-cost bitset*, not only its minimum. Every decoded route was checked for visiting each city once, closing at 0 and attaining the reported cost.

20.2 Reproducibility and scope

The package contains `axis0_double_uu_reference.py` and `verification.json`. The source SHA-256 recorded in that JSON is

```
aa29645d6c1020fa7c9d740a652c7cd4
e38be757452f40baf640b488ae3dd66a
```

The tests establish the reported finite-case behavior; the general guarantees rely on the proofs above. They do not measure hardware speed or establish all conceivable field profiles. The prior version's 4,781 golden-ratio recurrence checks are a separate historical record and were not rerun as part of this suite. [V2, p. 20; current execution record]

21 Migration from version 2

REVISION LEDGER / PRESERVED DEFINITIONS AND EXPLICIT REFRAMINGS

Component	Retained	Version 3 enhancement
Axis 0	ALL admitted operator-fields.	Exact mask/operator correspondence and an explicit capacity account.
UU	Named operator, previously without an assigned law.	Boolean double-UU paired carrier, following the new author-issued statement.
SDF	Geometry, metric, zero set and local signed evaluation.	Boolean phase view plus a declared reconstruction; a sign bit is not mistaken for distance magnitude.
PSI, jitter, φ	Branching, parity, lowercase scaling and profile-specific transition laws.	Typed connection to phase or reachability kernels; roles are not conflated.
Klein bifurcation	Local twisted-distance carrier and two addressed ports.	Phase/mask transformation laws and a discrete seam example.
SDF = T	Signed geometric field-time in normalized units.	Separate arrival-time/history reading and two distinct Now interfaces.
Grounding	An assigned observable and exact predicate.	Empty-frontier completion for profile R, with its own proof rather than a thermodynamic inference.
Routing	No complete route-solving profile in V2.	Proven unit-grid routing and an optional exact integer-cost tour extension.

21.1 What remains a parameter or a further construction

A full universal evaluator, a self-copying system, arbitrary weighted/time-dependent spatial motion, physical causal laws, a device-level implementation, cryptographic security and the unresolved source-specific operator names remain outside the implemented profiles. They have not been silently assigned meanings from their names or from the generated responses.

Consolidated formal statement

Tom Klootwijk's Axis 0 definition is realized here as a field-represented operator domain with Boolean double-UU state/action views. Spatial shifts transport fields; XOR applies flips; OR preserves alternative feasibility. An exact SDF interpretation requires its metric and interface, and an exact time or tour interpretation requires its declared decoder. Under the stated profiles, the resulting equations support provable reachability and exact integer-cost tour computation with an explicit state-space and resource account.

22 Source lineage and references

DOCUMENT RECORD / PRIMARY CONTEXT AND REPRODUCIBLE CONSTRUCTIONS

S1. Original six-page source conversation. Uploaded PDF whose filename begins *men T room a circle a sphere a cone an apex legenda*. Page 2 adds jitter and branching terminology; page 5 contains the author's ALL-operators definition of Axis 0. Author prompts and generated responses are separate source voices.

S2. Eight-page continuation. Uploaded hyphenated-filename PDF beginning *men-T-room-a-circle-a-sphere-a-cone-an-apex-legenda*. Pages 6–8 contain the seed/grounding/readout discussion, SDF = T, Present, entropy and Now. The document contains generated explanatory responses.

S3. New six-page source conversation. Uploaded PDF beginning *ladybug-anti-violence-tile-in-zaandam-at-the-train-station*. Author entries are visible at 07:33 on page 1, 07:38 on page 2 and 07:44 on page 4. The Boolean-layer equation is on page 3; the instantaneous TSP claim is on pages 2 and 4; the five applications are on pages 5–6.

V2. Prior formalization. *Tom Klootwijk: Axis 0, Operator–SDF–T framework, enhanced formal specification, version 2.0*, 21 pages, delivered earlier in this conversation. It is an earlier constructed specification, not independent experimental evidence for claims in S1–S3.

R1. John C. Hart (1996). *Sphere tracing: A geometric method for the antialiased ray tracing of implicit surfaces*. *The Visual Computer* 12(10), 527–545. DOI: 10.1007/s003710050084. Author-institution publication record consulted for the geometric-distance/SDF context.

R2. J. A. Sethian (1996). *A fast marching level set method for monotonically advancing fronts*. *PNAS* 93(4), 1591–1595. DOI: 10.1073/pnas.93.4.1591. Author-hosted manuscript dated October 1995 consulted for the stated front/arrival-time model.

R3. Michael Held and Richard M. Karp (1962). *A dynamic programming approach to sequencing problems*. *Journal of the Society for Industrial and Applied Mathematics* 10(1), 196–210. DOI: 10.1137/0110015. Primary publication reference for subset-state sequencing; the cost-bit construction in this revision is stated and proved here.

R4. University of Waterloo TSP research site. *Solving a TSP: Cutting Planes*. Research-team presentation of the Dantzig–Fulkerson–Johnson formulation, degree conditions and subtour elimination. Consulted for the exact tour constraint distinction.

R5. G. Edward Suh and Srinivas Devadas (2007). *Physical Unclonable Functions for Device Authentication and Secret Key Generation*. Design Automation Conference. Author-hosted primary paper; physical chip variation is distinct from deterministic Boolean pattern complexity.

R6. Python Software Foundation. *Built-in Types: Bitwise Operations on Integer Types*. Official documentation consulted for integer shifts, XOR, complement and bit indexing conventions.

External references were consulted on 18 September 2026. They supply the explicitly cited technical context; they are not presented as endorsements of the Axis 0 definition or as proofs of the proposed profiles. All new numerical examples, proofs and tests are identified within this revision.